

Midterm Review

All topics of Midterm



CSE110 | Programming Language I

NOTE: No Submission for this LAB

CODING TASKS

Task: 01

After working for a year finally Karim got his salary and bonus for Eid 2026, now he decides to do some kind of shopping for his family. His salary is x Tk and he got 75% bonus of his salary. He decides to spend money on shopping precisely so that everyone gets what they want. He planned to distribute 45% of his total money for his wife and 45% for his son. The remaining 10% will be saved for emergency cases. His wife wants a new laptop that costs around 90,000 Tk and his son wants a PS5 that costs about 70,000 Tk with games and an extra controller to play with friends that costs about 7,000 Tk. Now, take x as input from the user and see if Karim can fully fulfill his families wish, if not possible then see whether he can use remaining budget from son and wife to fulfill each other's wish. The saving part should not be touched by any means.

Sample Input	Sample Output
Enter Karim's Salary (x): 110000	Amount with Bonus: 192500.00 Wife Budget: 86625.00 Son Budget: 86625.00 Savings: 19250.00 Shopping Done for Wife with remaining money from Sons budget! Shopping Done for Son!
Enter Karim's Salary (x): 100000	Amount with Bonus: 175000.00 Wife Budget: 78750.00 Son Budget: 78750.00 Savings: 17500.00 Cannot do shopping for wife even with remaining money from his son's budget! Shopping Done for Son!

Task: 02

A group of astronauts established a space colony on Mars in 1999. Due to the harsh conditions, they must import essential supplies from Earth. The cost of these supplies has increased each month due to inflation and rising transportation expenses. The price of oxygen cylinders has increased by $p\%$ each month based on the previous month's price. The price of water reserves has increased by $q\%$ each month based on the previous month's price. Write a program that takes as input the price of an oxygen cylinder and a unit of water reserve in 1999, along with the values of p and q . The program should then calculate and print the current price of both items and determine which item's price has increased more in total over these years. Assume, the current year is 2024.

Sample Input	Sample Output
Enter the initial price of an oxygen cylinder in 1999: 120 Enter the initial price of a unit of water reserve in 1999: 20 Enter the monthly inflation rate (percentage) for oxygen cylinders: 1 Enter the monthly inflation rate (percentage) for water reserves: 1.5	Current price of an oxygen cylinder: 2675.78 Taka Current price of a unit of water reserve: 2081.78 Taka The price of oxygen cylinders has increased more.
Enter the initial price of an oxygen cylinder in 1999: 45 Enter the initial price of a unit of water reserve in 1999: 15 Enter the monthly inflation rate (percentage) for oxygen cylinders: 1 Enter the monthly inflation rate (percentage) for water reserves: 2.3	Current price of an oxygen cylinder: 1003.42 Taka Current price of a unit of water reserve: 18083.77 Taka The price of water reserves has increased more.

Task: 03

Write a Java program that takes two integer numbers: a & b as user inputs and calculates $a\%b$ and a/b without using the modulus (%) and division(/) operators respectively.

Task: 04

Write a Java program that takes three integers: U, L and D. Here, U & L denotes the upper & lower limit respectively and D is a specific digit from 0 to 9. Your task is to find out which numbers from U to L (both inclusive) contain the digit D.

Sample Input	Sample Output
20 50 3	23, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 43.
15 29 7	17, 27.

Task: 05

You are developing a **banking app** that allows users to input their PIN code for authentication. To improve security, the app records the frequency of each digit used in the user's PIN code over time. The app will then display the frequency of each digit (0-9) in the PIN code entered by the user.

Input	Output
Enter your PIN code: 445899	Frequency of digits: 0: 0 1: 0 2: 0 3: 0 4: 2 5: 1 6: 0 7: 0 8: 1 9: 2

Task: 06

At the Space Navigation Center, logs are constantly transmitted as integers, but due to a sensor malfunction, data arrives in two mixed formats: pure binary codes representing navigation vectors and standard decimal numbers representing fuel percentages. Officer Vance needs a single automated program to process these incoming signals without manual mode selection. If the transmitted number consists exclusively of the digits 0 and 1 (like 1011 or 0), the system must immediately identify it as a binary vector and output its decimal value. However, if the signal contains any digit from 2 through 9 (like 45 or 1020), the system must recognize it as a standard decimal value and convert it into binary so the flight computer can read it.

Write a Java program that reads a single positive integer n from the console and automatically determines the correct conversion.

Task: 07

The archeologist disarms the outer door, but a second chamber door requires an exact geometric measurement to slide open. The lock displays a target area of 10, which requires a precise length answer of approximately 3.162 rather than a simple whole number. The temple's magical field still blocks advanced math utilities, so you must extend your search past whole numbers into fractional increments to compute the accurate decimal square root using basic arithmetic.

Write a Java program that reads a positive integer n and computes its square root including decimal places up to a target precision (e.g., 3 decimal places) using only basic loop operations.

[Do not use `Math.sqrt()` or any math library functions. Process everything using numerical loops and simple arithmetic (+, -, *, /)]